

# What is the deal with the "AI evaporates 80% of water" argument?

Posted by u/a\_potato\_ate\_me • • 0 points • 23 comments

I'm not here to argue about AI itself, I'm just genuinely confused why this is such a common argument. We all learned the water cycle, yeah? And that whole "Matter cannot be created or destroyed" thing? It was explained how rain forms?

A good example of what I'm talking about that I can think of is coffee makers. They work by you filling it with water, then they heat up to where the water evaporates into steam, that steam then condensates on the lid of the coffee maker and drips through the coffee and filter into the pot.

Is there something after the evaporation that makes the water totally unsafe to use that I'm missing? Did they learn how to make water vanish? I'm genuinely confused on this argument.

## Comments

u/Any-Mathematician946 • 1 points • 2026-01-23 00:27 UTC

Learn about aquifers and you will be horrified by why this is bad.

u/Sad-Pattern-1269 • 1 points • 2026-01-22 01:37 UTC

when you draw water from the water table faster than it replenishes, the earth compacts and makes it far harder for it to refill. This is a problem of all industrial water consumption, this is just one example.

Additionally water might not end up where we want it, rain over the ocean for example is 'wasted'

u/Underhill42 • 1 points • 2026-01-20 22:00 UTC

It doesn't disappear completely - but it's not being put back into the river where the people downstream can use it again. THAT'S the problem - there's far more than enough water on Earth, but that doesn't actually do anyone any good. The limiting factor on a population is usually how much fresh water is available locally.

If you drink or wash with the water it ends up in the sewer where it's treated and the water is dumped back into the river for the next town down stream to use. (Get over it, humanity has been drinking recycled dinosaur fluids for our entire existence.)

Evaporate it though and it's carried away by the winds, likely to rain down somewhere half a continent away where it's useless to your local communities.

u/Great-Powerful-Talia • 0 points • 2026-01-20 05:36 UTC

A big motivator for the more outlandish arguments is that the AI experience sucks and people really want to come up with *\*any\** reason that it's a moral imperative to oppose it.

The big warning sign for this is that they *\*want\** bad things to be happening, and get upset to discover that they aren't. (You see this a lot in, like, Pizzagate conspiracies- people actively hoping that the Dems are a pedo ring because they want a reason to hate the Dems.)

As a result, everything is exaggerated wildly.

Suddenly, instead of AI art looking like shit and being everywhere (not strictly true in all cases, but you have to really look to find anything interesting because it's so easy to spam bad AI art), it lacks a human soul and is therefore blasphemous.

AI water usage is a lot, but it's not really comparable to golf- it's reasonable to go for something that you can oppose, since banning golf isn't viable, but the choice is often motivated by a desire to hate AI. And that's motivated by the fact that AI's annoying as hell and increasingly appears to be completely useless.

u/GlobalWatts • 2 points • 2026-01-20 05:04 UTC

If you learned about the water cycle, then surely you also remember the part about how it takes tonnes of effort and resources to make it both safe to drink and available where it's needed, for which we only have a finite capacity. We don't drink straight rainwater. That's the main problem with "using" water, whether you're watering your lawn or cooling a data center that's partly used for running "AI" (whatever that term means this week).

The fact that evaporated water will eventually rain down somewhere doesn't change that. *\*Eventually\** and *\*somewhere\** don't help people who need water *\*here\** and *\*now\**. You can absolutely use water faster than the water cycle can replenish it, and you can process and transport it so people can drink it. And there are real concerns with data centers using up fresh water sources, making it unavailable for higher priority needs; especially since tech companies aren't always shouldering the costs of improving infrastructure accordingly.

It's weird how a thing everyone understood as ten year olds is now genuinely confusing to grown-ass adults. It's like people are going out of their way to be ignorant.

It's even weirder how this become one of the main talking points against. As if there weren't already a hundred other massive problems with it.

Now if you want to argue how much water is used for "AI" compared to other major water consumers, and whether the benefits we are supposedly getting are really worth it, that's not invalid but it's also not an argument with a clear, objective answer.

u/coanbu • 0 points • 2026-01-20 01:58 UTC

I have not heard this 80% argument. What is it even referring to?

Water usage is overblown in some circles (and underplayed in others) but can be a considerable issue

on a local level depending on the context. Power usage is much more concerning particularly if it keeps scaling up like the boosters are predicting.

u/New\_Line4049 • 2 points • 2026-01-20 00:39 UTC

Oh, sure, the water comes back down as rain, but nobody wants to drink rainwater, so you're turning clean, drinkable fresh water into dirty undrinkable water that needs to be processed again if you want to drink it.

Its also environmentally harmful, water in the atmosphere acts like a greenhouse gas so shoving loads more into the atmosphere is not a great plan.

u/Fearlessleader85 • 4 points • 2026-01-20 00:33 UTC

AI doesn't do anything truly different from any other server farm. They all require cooling, and generally, that cooling is ultimately done by using large AC units imaginatively called "chillers" to chill some water or sometimes another working fluid that's circulated through the equipment and either used to cool air which is then directed over the equipment, or some might directly cool the equipment. The heat the chillers remove from that chilled water can be either directly rejected to the air through large radiators, or it can be put into some other water that then is run through cooling towers where a portion of it is evaporated off to cool the rest, which is recirculated back to the chillers. Using the water is MUCH more efficient, so that's more common.

The reason "80%" of the water is evaporated is because as the water evaporates, it leaves behind any solutes, so the remaining water has higher levels of concentration of solutes. These solutes eventually will reach a concentration high enough that they start coating the heat exchangers, ruining the equipment. So, to combat that they dilute it by pouring a portion of the high solute water down the drain and adding clean, low solute makeup water to replace both that and the water that evaporates. So, 100% of the water is "lost". None of it is lost forever. It will show up as liquid water somewhere.

The only reason AI is of any special concern is the scale. It is immensely processor intensive, producing a lot of heat and then using a bunch of water. And the issue with that water use isn't just a general problem. Much of the world has absolutely no problem with water availability. The issue is extremely location specific.

These big data centers are MASSIVE, and what do you do when you want to build a huge facility that only needs a handful of people to work in it? You find somewhere with loads of cheap land near a small town or city. Somewhere like Prineville, Oregon, where Meta and some others have set up huge data centers. Now, if you look up some pictures of Prineville, you may notice that it's not exactly a lush, green paradise. It is full desert, getting 9-12" of rain per year.

So, these facilities require a massive amount of water are put in where it almost doesn't rain, where is it coming from? Well, they can start pulling it from the river, but then they're fighting with both local agricultural irrigation and the need for the river to not literally run dry without completely destroying the ecosystem of everything downstream. Or, they can pull it from the aquifers using huge wells. Water cannot filter into the aquifers as fast as these wells can pull it out, so they actually drop the water table, leaving other wells in the area dry. And dry aquifers have a nasty habit of collapsing, meaning they actually can't refill, not to the point that they once could hold. This causes a bunch of short term problems, but the real problems are long term. Eventually, these areas might end up so dry you can't practically dig a well deep enough to get water.

So, an AI data center on Windward O'ahu would be extremely expensive to build and ecologically problematic for some other reasons, but at over 250" of rain per year and a small population center, water isn't an issue. AI data centers built in the desert or the water starved ag land of the California

central valley are extremely problematic, but only produce local problems. That doesn't mean they don't matter, but it's not like the whole world is going to run out of fresh water. It isn't. It literally can't except through pollution bad enough that the aquifers are polluted and rain is undrinkable. That is still a possible concern, but we actually have been doing a little better on that front in recent decades. At least until very recent lifting of many regulations in one of the worlds' largest polluting countries.

**u/frogonthefloor** • 1 points • 2026-01-29 23:57 UTC

amazing comment, thanks for explaining all of this

**u/mredding** • 2 points • 2026-01-19 22:12 UTC

The Colorado river - one of the largest rivers in the world, hasn't reached the Gulf of California since 1998. The United States consumes almost all its water. The last stop is the Morelos Dam, in Mexico, where they divert the last of it for irrigation.

All this is to say, you can use water faster than you can replace it.

**u/azuth89** • 6 points • 2026-01-19 21:33 UTC

The water cycle does not mean it's going to come back down somewhere accessible or useful to the places where it was extracted.

Particularly in places where water resources are often stretched already, this is an issue.

It isn't everywhere, but its true of many places they're building new centers.

The other issue is the type of water used, often theyre pulling from municipal supplies which incurs a lot of extra processing costs on local infrastructure and means they're competing not just for accessible water in general but for treatment capacity.

**u/archpaw**n • 1 points • 2026-01-19 21:13 UTC

Rain falls at a limited rate. Yes it's a lot, but agriculture uses a lot.

The mistake is thinking AI uses a lot. It sounds like a lot if you don't know how much agriculture uses, especially if you compare a datacenter or training a model to a single person. According to [here] (<https://www.instagram.com/p/DTbBEKJFMAY/>), it takes eight gallons of water for a one-minute video. I hate using an instagram comment as a source, so if anyone can find a better one, I beg you to post it. In any case, that sounds like a lot, but a cheeseburger takes 600 gallons. Nobody will make a big deal out of it if you eat one cheeseburger. And a minute of video is a lot. AI videos tend to be done in short clips, and making a minute worth of it isn't something a normal person is going to do for their own amusement.

**u/vlegionv** • 6 points • 2026-01-20 00:31 UTC

Hank green put up a really nuanced and well informed 23 minute long video about it. [[https://www.youtube.com/watch?v=H\\_c6MWk7PQc](https://www.youtube.com/watch?v=H_c6MWk7PQc)]([https://www.youtube.com/watch?v=H\\_c6MWk7PQc](https://www.youtube.com/watch?v=H_c6MWk7PQc))

tl;dr mf'ers on both sides are crazy.

**u/Reasonable\_Place\_481** • 4 points • 2026-01-19 21:20 UTC

But if you are not in an agricultural area, and the data center will be using 50% of the water the entire city already uses maxing out the estimated capacity of the water plant while not contributing to the cost to build a new plant, the yes it's an issue. Add to that the power demand with the same capacity constraints, residents have concerns.

**u/Runiat** • 3 points • 2026-01-19 20:52 UTC

>Is there something after the evaporation that makes the water totally unsafe to use that I'm missing?

You're missing the bit where 70% of it ends up falling into the ocean, making it fatally unhealthy for both humans and almost all the plants we rely on for food.

**u/Nuts4WrestlingButts** • 2 points • 2026-01-19 20:42 UTC

Clean, potable water is time and energy intensive to make. Data centers are popping up all over the place in areas without the infrastructure to support it so all the local citizens have their water supply being stolen.

**u/rhomboidus** • 2 points • 2026-01-19 20:39 UTC

Making drinking water costs money and energy. Lots of both.

Businesses building facilities that use tons and tons of drinking water puts a large strain on municipal water systems.

**u/bangbangracer** • 2 points • 2026-01-19 20:39 UTC

The big thing is that the water being pulled out of reservoirs doesn't immediately go back to the reservoir. The water cycle isn't very fast and the way we are using water depletes reservoirs faster than it's getting replenished already, and climate change isn't helping things. The problem is only going to get worse when we introduce a ton of new AI focused data centers.

**u/Justsomedudeonthenet** • 2 points • 2026-01-19 20:38 UTC

The big deal is that the water cycle we learnt about didn't talk about purifying water into clean drinking water.

Datacenters are using drinking water, whether from city supply or water from wells or aquifers. That puts a huge demand on those systems - enough that they might not be able to continue providing enough water for people to use.

The water isn't destroyed. It'll rain somewhere. But it takes tons of time for aquifers to naturally form and refill, and lots of power and equipment for us to purify water with treatment plants. That's the big problem.

**u/MarsThrow** • 2 points • 2026-01-19 20:37 UTC

The water cycle takes a long time, the books certainly make it feel like "oh it rains and everything is back to normal" but the water needs to work its way through the ground to where it needs to go. Additionally ground water was already being used up past replenishment rates in most areas, AI is using up way more water and electricity than our grids are meant for.

**u/Crafty-Money8507** • 26 points • 2026-01-19 20:37 UTC

The issue isn't that the water disappears forever, it's that data centers use massive amounts of clean freshwater for cooling and then release it as hot vapor into the atmosphere. Sure it eventually comes back as rain somewhere else, but that "somewhere else" might be hundreds of miles away

It's like if your neighbor kept borrowing your garden hose to fill their pool then just dumped all the water on their lawn - technically the water's still "there" but now you don't have any for your garden

**u/Special-Audience-426** • 4 points • 2026-01-19 20:49 UTC

So perhaps data centres should be building massive cooling towers like power stations used to have so they can recycle the water.

Or put data centres in the arctic and just leave the windows open.

**u/pdpi** • 5 points • 2026-01-19 21:31 UTC

> So perhaps data centres should be building massive cooling towers like power stations used to have so they can recycle the water.

They already have similar things to that. The vast majority of water in data centres is cycled back, and current consumption levels already factor in all the water that gets reused.

> Or put data centres in the arctic and just leave the windows open.

At least do it in Antarctica so there's actual land to build on. Also, good luck hiring local talent :)

**u/fixermark** • 2 points • 2026-01-20 14:58 UTC

They've actually built datacenters in countries like Iceland where they plumb the waste heat from the datacenter into the municipal heating system to take some of the strain off the town's primary heat generators.

**u/Runiat** • 6 points • 2026-01-19 20:54 UTC

>So perhaps data centres should be building massive cooling towers like power stations used to have so they can recycle the water.

If you're thinking of cooling towers similar to the ones the powerplant in The Simpsons has, those evaporate water to reuse other water, same as what OP is talking about.